

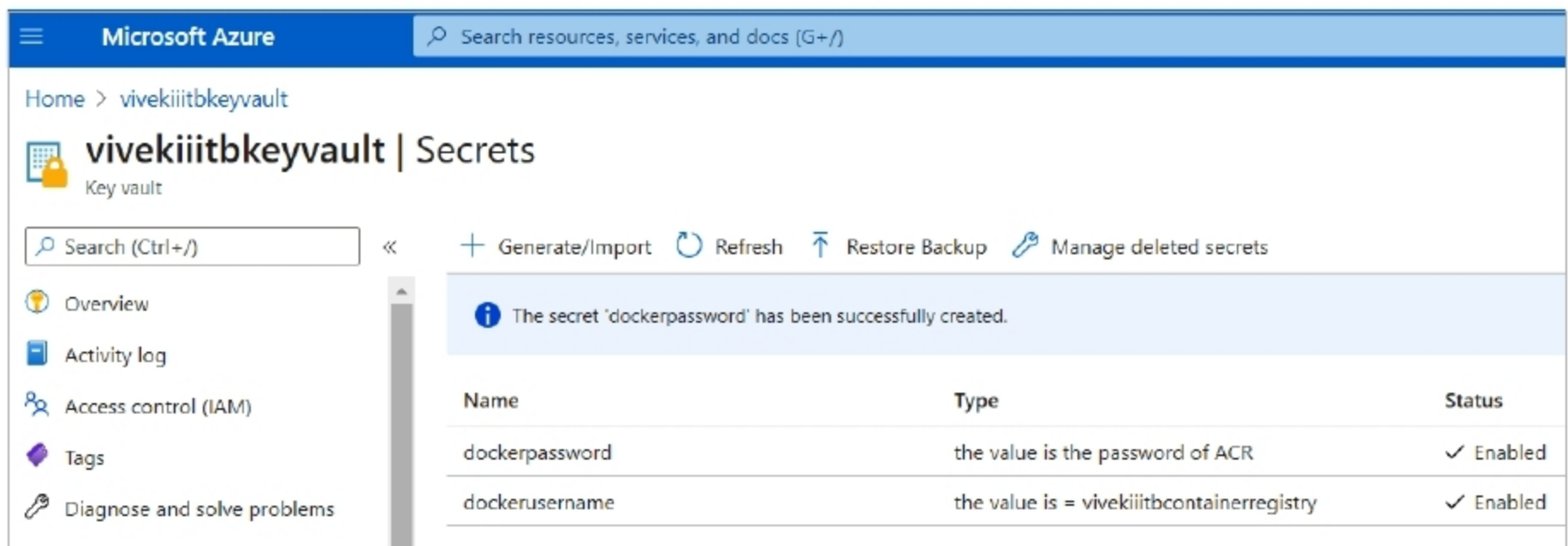
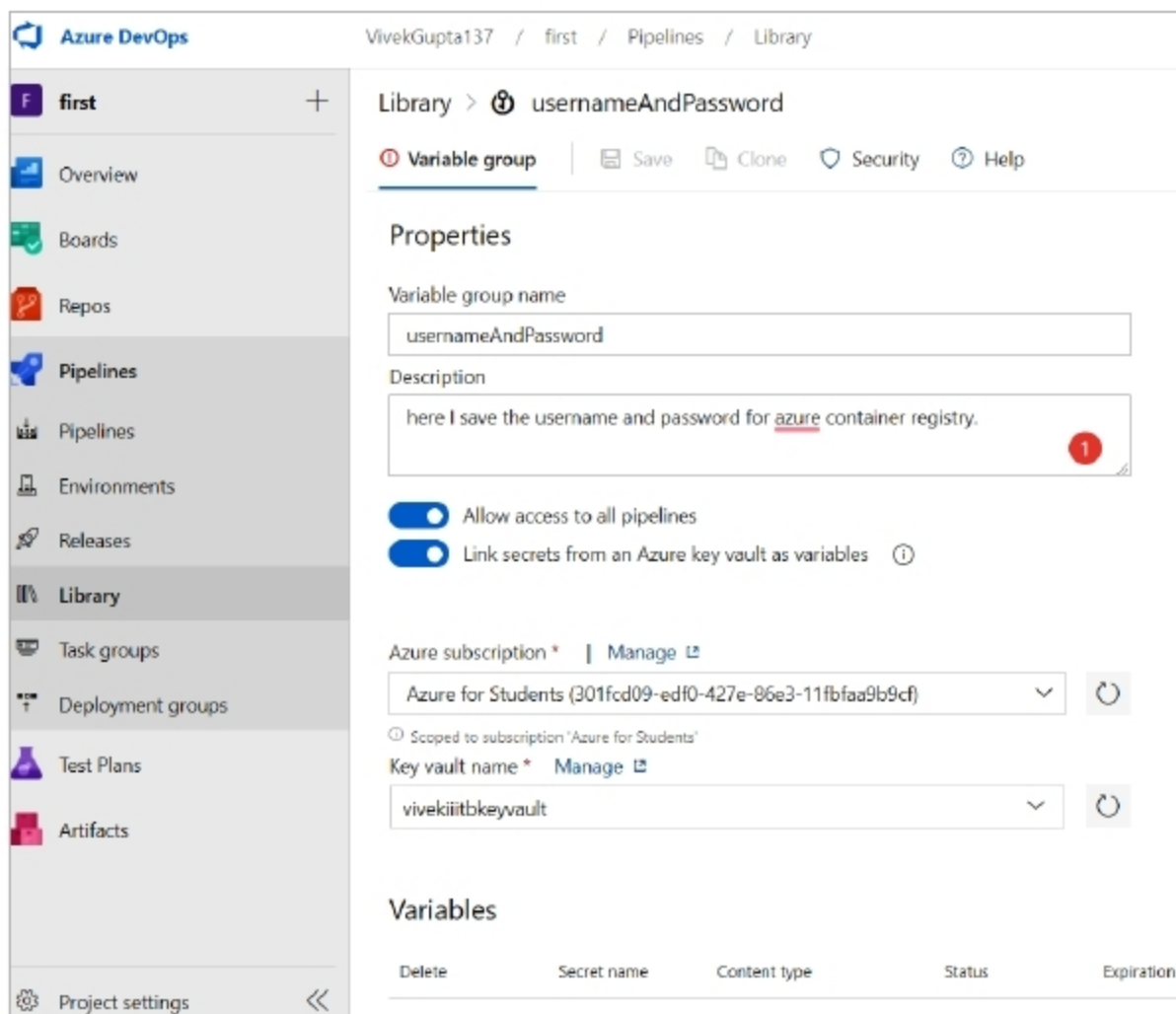
Figure 10: Secrets created in *vivekiiitbkeyvault* key vault

Figure 11: Creating the variable group

- Now you can navigate to the pipelines tab, and then enter *Create pipeline*. After this, it will ask you to connect to the Azure Repos Git repository, as shown in Figure 4.
- In the next step, select the Git repository and then choose the starter pipeline template to configure the pipeline, as shown in Figure 5.
- Now fill in the required details in the *azure-pipelines.yaml* file, as shown in Figure 6.

Using this YAML file, build the Docker image, and then push it into the Azure container registry.

- Lines 1-2:** These lines specify the pool name, which will run this pipeline.
- Line 4:** This specifies the sequence of operations that you would like to perform.
- Line 6:** As you would want to run command-line tasks, they are specified within scripts.

- Line 7:** This gives the *docker build* command to build the image.
 - Lines 8-9:** The image built previously is tagged, followed by the list of all the images.
 - Lines 10-12:** You are logged in to the Azure Container Registry by providing the user name and password of your resource. Push this image to the ACR (Azure Container Registry).
- Then click *Save and run*. You will see that the pipeline is queued, which is shown in Figure 7.
 - When it finishes its execution, you can see the logs by clicking on the *Job* menu, as depicted in Figure 8.

Enhancing the pipeline

As you can see, the user name and password in the YAML file are hardcoded so that one cannot reuse or update the password when it is needed from a central location. This also creates a security concern as the password will go into the source code. To overcome these problems, variables and Azure key vault are used.

Creating the Azure key vault resource

- Log in with the Azure subscription at <http://shell.azure.com/>.
- Create the resource Azure key vault in the resource group created earlier using the